Notes: Adelino, Ferreira, Giannetti, and Pires (RFS, 2023)

Trade Credit and the Transmission of Unconventional Monetary Policy

Contents

1 Big picture

- **Question.** How does unconventional monetary policy that purchases corporate bonds transmit to firms that do not issue eligible bonds and do not directly benefit from central bank purchases?
- **Policy shock.** The paper studies the European Central Bank's Corporate Sector Purchase Programme (CSPP), announced in March 2016 and implemented from June 2016, which made investment-grade bonds of euro-area nonfinancial firms eligible for purchase.
- **Main channel.** Eligible bond issuers experience cheaper access to public debt markets and then pass part of this liquidity to their customers through trade credit. In this sense, large firms become private financial intermediaries inside production networks.
- **Direct supplier effect.** Firms with CSPP-eligible bonds increase accounts receivable relative to noneligible firms after the policy. Accounts receivable are interpreted as trade credit extended to customers.
- **Customer effect.** Customers of eligible suppliers increase accounts payable relative to other customers after the CSPP. This is the mirror outcome: customers receive more trade financing.
- **Regional transmission.** Eligible firms are concentrated in core euro-area countries. Trade credit allows the liquidity injected into core firms to propagate to customers in periphery countries.
- **Financial-constraint transmission.** The increase in accounts payable is stronger for customers that are more financially constrained, such as unrated, non-investment-grade, high-liability, or low-tangible-asset firms.
- **Real effects.** Customers of eligible suppliers raise investment, inventories, accounts receivable, and employment. Thus the trade-credit channel affects real activity, not only balance-sheet composition.
- **Market-structure effect.** Eligible firms expand or retain customer relationships and gain market share, especially in core countries. The policy may therefore increase product-market concentration upstream.
- **Takeaway.** Corporate bond purchase programs can redistribute monetary stimulus across firms and regions through production networks, but they can also strengthen the competitive position of firms directly targeted by the policy.

2 Data and measurement

2.1 The Corporate Sector Purchase Programme

- The CSPP is an ECB large-scale asset purchase program targeting corporate bonds.
- Eligibility requires that bonds are issued by nonfinancial firms in the euro area and have investment-grade ratings from at least one accepted rating agency.
- The CSPP was announced in March 2016 and purchases began in June 2016.
- The program lowers yields and increases bond issuance opportunities for eligible firms.
- Eligible firms are overwhelmingly large, publicly connected to capital markets, and concentrated in core euro-area countries.

Interpretation: The policy directly targets public corporate bond markets, not bank loans or trade credit. The paper's contribution is to show how liquidity nevertheless reaches smaller firms through customer-supplier links.

2.2 Firm sample

- The main firm-level data come from Bureau van Dijk's Orbis.
- The sample covers nonfinancial firms in euro-area countries over 2013–2017.
- Financial firms and public administration entities are excluded.
- Small companies, as classified by Orbis, are excluded.
- Firms must have nonmissing values for the main outcomes: accounts receivable to sales and accounts payable to sales.
- The final sample contains 510,298 unique firms and 2,248,514 firm-year observations.

Interpretation: The sample is broad and includes many private firms, allowing the paper to study firms that do not issue bonds and would be missed in public-firm datasets.

2.3 Customer-supplier network data

- The paper uses FactSet Revere to identify customer-supplier links.
- Links include direct relationships disclosed by suppliers and reverse relationships disclosed by customers.
- A customer is considered exposed if it has at least one CSPP-eligible supplier.
- A continuous exposure measure is the share of the customer's suppliers that are eligible.
- The network data allow the authors to connect eligible bond issuers to downstream customers and to study both sides of the trade-credit relation.

Interpretation: The network data transform the paper from a study of eligible bond issuers into a study of monetary-policy propagation through production chains.

2.4 Main variables

- **Eligible:** dummy equal to one if a firm had bonds eligible for CSPP purchases before the announcement.
- **Post:** dummy equal to one for the post-CSPP period.
- **Accounts receivable:** accounts receivable scaled by sales; interpreted as trade credit extended by a firm to customers.
- **Accounts payable:** accounts payable scaled by sales; interpreted as trade credit received from suppliers.
- **Has eligible supplier:** dummy equal to one if a firm has at least one eligible supplier.
- **Eligible suppliers share:** fraction of the firm's suppliers that are eligible.

Interpretation: The supplier-side outcome is accounts receivable; the customer-side outcome is accounts payable. The symmetry of these outcomes is essential to the trade-credit interpretation.

2.5 Control variables and additional outcomes

- Controls include firm size, cash, property, plant, and equipment, net margin, and liabilities.
- Investment outcomes include asset growth, capital expenditures, inventories, changes in accounts receivable, and labor growth.
- Financing outcomes include changes in total debt, long-term debt, short-term debt, accounts payable, and cash.

- Product-market outcomes include the number of customer relationships maintained, the number of new customer relationships, and sales market share at the four-digit SIC level.

Interpretation: The additional outcomes distinguish balance-sheet rearrangement from real economic effects and product-market consequences.

3 Models and empirical specifications

3.1 Supplier-side baseline difference-in-differences

The supplier-side regression studies accounts receivable of eligible bond issuers:

$$AR_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \beta(Eligible_i \times Post_t) + \Gamma' X_{i,t-1} + \varepsilon_{i,t}, \tag{1}$$

where $AR_{i,t}$ is accounts receivable scaled by sales, α_i are firm fixed effects, $\lambda_{j,t}$ are industry-year fixed effects, $\gamma_{c,t}$ are country-year fixed effects when included, and $X_{i,t-1}$ are lagged firm controls.

Interpretation: The coefficient β measures whether CSPP-eligible firms extend more trade credit after the policy relative to noneligible firms, after absorbing firm, industry-year, and country-year factors.

3.2 Event-year version

The paper also estimates year-specific treatment effects:

$$AR_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \sum_{\tau} \beta_\tau (Eligible_i \times 1\{Year = \tau\}) + \Gamma' X_{i,t-1} + \varepsilon_{i,t}. \tag{2}$$

Interpretation: The year-specific coefficients show whether treated and control firms were already trending differently before the CSPP. Insignificant pre-period coefficients support parallel trends.

3.3 Customer-side baseline difference-in-differences

The customer-side regression studies accounts payable of firms with eligible suppliers:

$$AP_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \beta(HasEligibleSupplier_i \times Post_t) + \Gamma' X_{i,t-1} + u_{i,t}. \tag{3}$$

A continuous version replaces $HasEligibleSupplier_i$ with $EligibleSuppliersShare_i$:

$$AP_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \beta(EligibleSuppliersShare_i \times Post_t) + \Gamma' X_{i,t-1} + u_{i,t}. \tag{4}$$

Interpretation: If eligible suppliers extend additional trade credit, customers should report more accounts payable after the CSPP.

3.4 Investment-grade cutoff tests

To sharpen identification, the paper compares eligible and noneligible issuers near the investment-grade boundary:

$$AR_{i,t} = \alpha_i + \lambda_{j,t} + \beta(Eligible_i \times Post_t) + \Gamma' X_{i,t-1} + \varepsilon_{i,t} \tag{5}$$

within rating bandwidths such as BBB+ to BB- and A+ to B-.

Interpretation: Firms near the cutoff are more comparable. A large effect near the cutoff supports the view that eligibility, not broad firm quality, drives the trade-credit response.

3.5 Eligible issuers versus eligible nonissuers

The paper splits eligible firms into those that issue bonds after the CSPP and those that do not:

$$AR_{i,t} = \alpha_i + \lambda_{j,t} + \beta_1(EligibleIssuer_i \times Post_t) + \beta_2(EligibleNonissuer_i \times Post_t) + \Gamma' X_{i,t-1} + \varepsilon_{i,t}. \quad (6)$$

Interpretation: If the mechanism is lower external funding cost, the trade-credit increase should be strongest among eligible firms that actually issue bonds after the CSPP.

3.6 Financial-constraint heterogeneity

The paper tests whether the customer effect is stronger for constrained customers:

$$AP_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \beta(HasEligibleSupplier_i \times Post_t) + \Gamma' X_{i,t-1} + u_{i,t}, \quad (7)$$

run separately for subsamples defined by ratings, age, liabilities, tangible assets, and other proxies.

Interpretation: If trade credit relaxes financing constraints, the effect should be stronger where alternative finance is more limited.

3.7 Regional transmission

To test transmission from core to periphery, the paper estimates supplier and customer effects separately by region:

$$Outcome_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \beta(CoreEligibleExposure_i \times Post_t) + \Gamma' X_{i,t-1} + e_{i,t}. \quad (8)$$

Interpretation: Because eligible bond issuers are mostly in core countries, a customer effect in periphery countries means that monetary stimulus crosses borders through production networks.

3.8 Real effects and product-market outcomes

The real effects regressions use outcomes such as asset growth, CAPEX, inventories, labor growth, debt, cash, customer relationships, and market share:

$$Y_{i,t} = \alpha_i + \lambda_{j,t} + \gamma_{c,t} + \beta(Treatment_i \times Post_t) + \Gamma' X_{i,t-1} + \epsilon_{i,t}. \quad (9)$$

Interpretation: These regressions ask whether trade credit changes only financial statement line items or also affects employment, investment, customer acquisition, and market power.

4 Identification and empirical design

4.1 Policy-based treatment variation

- Eligibility is based on preexisting bonds satisfying CSPP criteria.
- The policy shock arrives in 2016 and changes the funding environment of eligible firms.
- Eligible firms are compared with noneligible firms before and after the CSPP.
- Customer exposure is determined by preexisting supplier relationships.

Interpretation: The main identifying assumption is that absent the CSPP, eligible firms and customers connected to eligible suppliers would have followed similar trade-credit trends to their controls.

4.2 Parallel-trends evidence

- Event-year estimates show no statistically significant pretreatment differences in accounts receivable.
- Customers of eligible suppliers also do not show strong pre-CSPP divergence in accounts payable.
- The treatment effects appear after the announcement and implementation of the CSPP.

Interpretation: The timing supports the interpretation that the CSPP caused the trade-credit changes, rather than preexisting trends.

4.3 Investment-grade cutoff

- Eligibility depends crucially on investment-grade ratings.
- The paper compares firms close to the investment-grade cutoff.
- The effect near the cutoff is large, about 0.27 to 0.28 in the tight BBB+ to BB- window.

Interpretation: This design reduces the concern that treated firms are simply much better or larger than controls. Near-cutoff comparisons make treatment and control firms more comparable.

4.4 Mechanism evidence

- Eligible firms that issue bonds after the CSPP increase accounts receivable more than eligible firms that do not issue.
- This supports the claim that the trade-credit effect is tied to lower external financing costs.
- Matching, size decile-year fixed effects, and country-year fixed effects reduce the risk that size, region, or macro shocks drive the estimates.

Interpretation: The mechanism is not merely that eligible firms are large. It is that eligible firms can tap the bond market more cheaply and redistribute liquidity to customers.

4.5 What the paper cannot fully prove

- Eligible firms are not randomly selected.
- Customer-supplier links in FactSet Revere cover material relationships and may miss some smaller links.
- Trade credit is measured from accounting data and cannot always identify contract terms directly.
- Some contemporaneous macro shocks could still interact with region, industry, or firm type despite extensive fixed effects.

Interpretation: The paper's credibility comes from multiple complementary tests rather than a single experiment.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- The sample contains 2,248,514 firm-year observations and 510,298 unique firms.
- Eligible firms are rare in the full Orbis sample because the sample includes many private and smaller non-bond-issuing firms.
- Mean accounts receivable is 0.3012 of sales, while mean accounts payable is 0.2284 of sales.
- Median assets are only about 3.3 million euros, confirming that the sample includes many firms far from public bond markets.

Economic message: The broad firm sample makes the paper especially suited to studying indirect transmission from bond-market policy to private firms.

Table 1
Summary statistics

	Mean	Median	SD	Minimum	Maximum	Number of observations
Eligible	0.0003	0.0000	0.0177	0.0000	1.0000	2,248,514
Has eligible supplier	0.0007	0.0000	0.0257	0.0000	1.0000	2,248,514
Eligible suppliers share	0.0002	0.0000	0.0125	0.0000	1.0000	2,248,514
Accounts receivable	0.3012	0.1849	0.5574	0.0000	4.6879	2,248,514
Accounts payable	0.2284	0.1134	0.5555	0.0000	4.9558	2,248,514
Assets (€million)	30.1234	3.2753	703.8566	0.0000	198,929	2,248,514
Sales (€million)	22.5520	3.1251	367.4128	0.0000	107,970	2,248,514
Cash	0.1130	0.0480	0.1536	0.0000	0.8158	2,248,514
PPE	0.2358	0.1303	0.2600	0.0000	0.9767	2,248,514
Net margin	−0.0528	0.0157	0.7375	−7.0694	1.7098	2,248,514
Liabilities	0.6402	0.6690	0.2958	0.0035	1.8202	2,248,514

This table shows mean, median, standard deviation, minimum, maximum, and the number of observations for each variable. Table A.1 in the appendix defines the variables. The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. Variables are winsorized at the top and bottom 1%.

5.2 Figure 1 and Figure 2: Where the CSPP liquidity enters

Read:

- Figure 1 shows that CSPP-eligible bonds are concentrated in core euro-area countries.
- Core countries represent a larger share of eligible bonds and eligible outstanding amounts than their share of euro-area GDP.
- Figure 2 shows that investment-grade bond issuance increases more strongly in core countries after the CSPP.
- Non-investment-grade issuance provides a comparison showing that the pattern is tied to the eligibility segment.

Economic message: The direct injection of liquidity is geographically concentrated. The key question is whether trade credit transmits this liquidity to customers elsewhere.

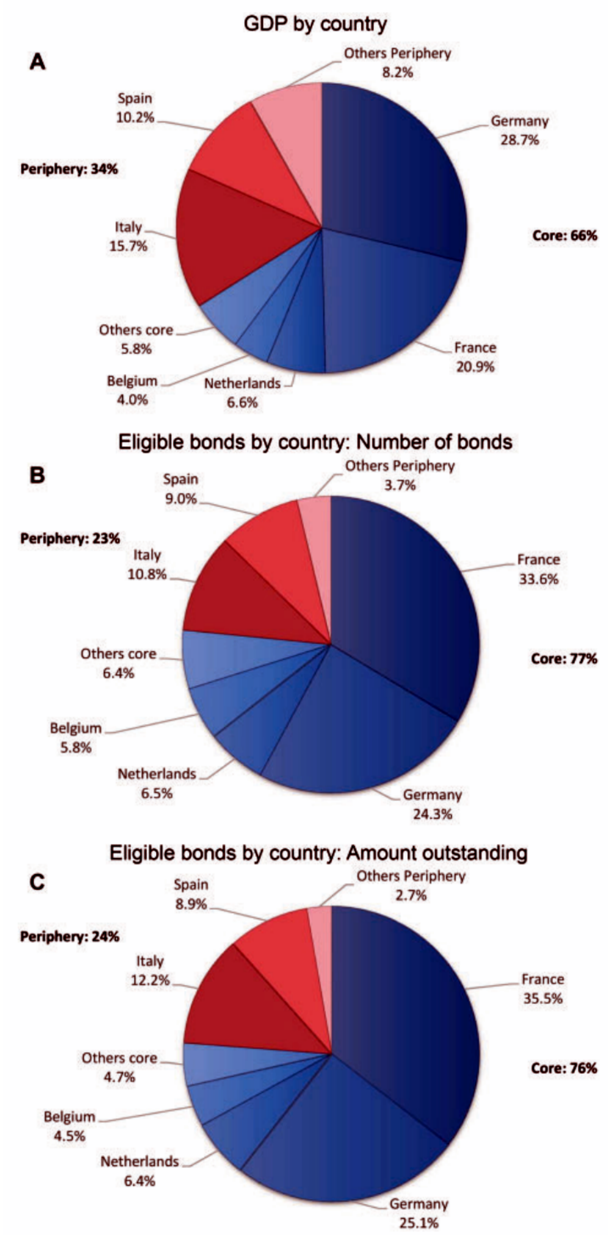

Figure 1: Eligible CSPP bonds by country

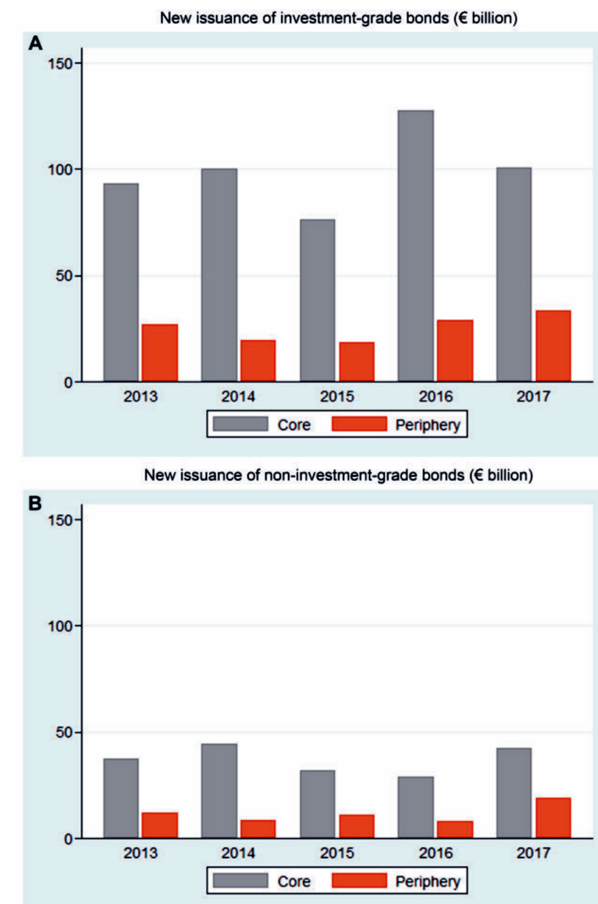

Figure 2: Bond issuance around the CSPP

5.3 Table 2: Eligible firms' accounts receivable

Read:

- Eligible firms increase accounts receivable to sales by about 10.3 percentage points after the CSPP in the baseline specification.
- With firm controls and country-year fixed effects, the coefficient remains large at 8.9 percentage points.
- Year interactions show no significant difference in 2014 or 2015, but significant increases in 2016 and 2017.
- A 10 percentage point increase corresponds to roughly 36 additional days receivable.

Economic message: Eligible firms use cheaper bond-market access to extend more trade credit to customers.

Table 2
Effect of CSPP on accounts receivable of eligible firms

	(1)	(2)	(3)	(4)	(5)	(6)
Eligible × Post	0.103***		0.102***		0.089***	
	(0.032)		(0.032)		(0.032)	
Eligible × 2014		−0.007		−0.007		−0.013
		(0.022)		(0.022)		(0.022)
Eligible × 2015		0.046		0.046		0.042
		(0.043)		(0.043)		(0.043)
Eligible × 2016		0.077**		0.077**		0.062*
		(0.038)		(0.038)		(0.038)
Eligible × 2017		0.156**		0.156**		0.139**
		(0.062)		(0.062)		(0.062)
log(Assets)			−0.013***	−0.013***	−0.014***	−0.014***
			(0.002)	(0.002)	(0.002)	(0.002)
Cash			−0.050***	−0.050***	−0.045***	−0.045***
			(0.005)	(0.005)	(0.005)	(0.005)
PPE			−0.086***	−0.086***	−0.087***	−0.087***
			(0.007)	(0.007)	(0.007)	(0.007)
Net margin			−0.006***	−0.006***	−0.006***	−0.006***
			(0.002)	(0.002)	(0.002)	(0.002)
Liabilities			−0.010**	−0.010**	−0.007	−0.007
			(0.005)	(0.005)	(0.005)	(0.005)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	No	No	Yes	Yes
Number of observations	2,248,514	2,248,514	2,248,514	2,248,514	2,248,514	2,248,514
R-squared	.74	.74	.74	.74	.74	.74

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts receivable to sales (*Accounts receivable*). *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

5.4 Table 3: Additional supplier-side tests

Read:

- Panel A excludes top-rated eligible firms. The treatment effect remains about 11.7 percentage points, showing that the result is not driven only by AAA-to-A firms.
- Panel B restricts the sample around the investment-grade cutoff. The effect is 0.273 to 0.282 in the BBB+ to BB- bandwidth and remains positive in the wider A+ to B- bandwidth.
- Panel C distinguishes eligible issuers from eligible nonissuers. Eligible issuers increase accounts receivable by about 13.3 percentage points, while eligible nonissuers do not show a significant increase.

Economic message: The mechanism is strongest where the policy changes funding access. Firms that actually issue bonds transmit more liquidity through trade credit.

Table 3
Effect of CSPP on accounts receivable of eligible firms: Additional tests

A. Estimates excluding top-rated eligible firms (AAA to A)

	(1)	(2)	(3)	(4)	(5)	(6)
Eligible × Post	0.117***		0.115***		0.104***	
	(0.039)		(0.039)		(0.039)	
Eligible × 2014		−0.007		−0.007		−0.012
		(0.026)		(0.026)		(0.026)
Eligible × 2015		0.011		0.011		0.007
		(0.033)		(0.033)		(0.033)
Eligible × 2016		0.091**		0.089*		0.076*
		(0.046)		(0.046)		(0.046)
Eligible × 2017		0.147**		0.145**		0.130*
		(0.067)		(0.067)		(0.067)
Controls	No	No	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	No	No	Yes	Yes
Number of observations	2,248,385	2,248,385	2,248,385	2,248,385	2,248,385	2,248,385
R-squared	.74	.74	.74	.74	.74	.74

B. Difference-in-differences with a bandwidth around the investment-grade cutoff

	BBB+ to BB-			A+ to B−		
	(1)	(2)	(3)	(4)	(5)	(6)
Eligible × Post	0.273***	0.282***	0.230**	0.187***	0.185***	0.114*
	(0.104)	(0.104)	(0.090)	(0.066)	(0.066)	(0.059)
Controls	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Number of observations	663	663	648	1,094	1,094	1,076
R-squared	.55	.56	.49	.57	.58	.57

Table 3, Panels A–B: rating and cutoff tests

Table 3
Continued

C. Eligible bond issuers versus eligible nonissuers

	(1)	(2)	(3)
Eligible issuer × Post	0.133***	0.133***	0.120***
	(0.044)	(0.044)	(0.044)
Eligible nonissuer × Post	0.031	0.029	0.017
	(0.027)	(0.027)	(0.027)
Controls	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes
Number of observations	2,248,514	2,248,514	2,248,514
R-squared	.74	.74	.74

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts receivable to sales (*Accounts receivable*). *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Panel A presents the results for eligible firms excluding top-rated (AAA to A) eligible firms. Panel B presents the results restricting the sample to a bandwidth around the investment-grade cutoff, either BBB+ to BB- in columns 1–3 or A+ to B− in columns 4–6. Panel C presents the results for eligible firms that have issued bonds and eligible firms that have not issued bonds after the CSPP announcement date. *Eligible issuer* is a dummy variable that takes the value of one if an eligible firm has issued bonds after the CSPP announcement date (i.e., in the period from March 2016 to December 2017), and zero otherwise. *Eligible nonissuer* is a dummy variable that takes the value of one if an eligible firm has not issued bonds after the CSPP announcement date, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

Table 3, Panel C: issuers versus nonissuers

5.5 Table 4: Customers' accounts payable

Read:

- Panel A shows that customers with at least one eligible supplier increase accounts payable to sales by 4.8 percentage points in the simplest specification and 3.2 percentage points with controls and country-year fixed effects.
- Panel B uses the share of eligible suppliers. A larger share of eligible suppliers is associated with a larger increase in accounts payable.
- The customer-side increase mirrors the supplier-side accounts-receivable increase.

Economic message: The customer results confirm that eligible firms are not only changing internal accounting; they are extending more financing to downstream firms.

Table 4
Effect of CSPP on accounts payable of eligible firms' customers

A. Eligible supplier dummy variable

	(1)	(2)	(3)	(4)	(5)	(6)
Has eligible supplier × Post	0.048***		0.045***		0.032*	
	(0.017)		(0.017)		(0.017)	
Has eligible supplier × 2014		0.030		0.028		0.025
		(0.023)		(0.023)		(0.023)
Has eligible supplier × 2015		0.039		0.038		0.039
		(0.027)		(0.027)		(0.027)
Has eligible supplier × 2016		0.034**		0.031*		0.018
		(0.016)		(0.017)		(0.017)
Has eligible supplier × 2017		0.110***		0.107***		0.092***
		(0.034)		(0.034)		(0.034)
log(Assets)			−0.045***	−0.045***	−0.046***	−0.046***
			(0.002)	(0.002)	(0.002)	(0.002)
Cash			0.022***	0.022***	0.026***	0.026***
			(0.005)	(0.005)	(0.005)	(0.005)
PPE			−0.048***	−0.048***	−0.047***	−0.047***
			(0.008)	(0.008)	(0.008)	(0.008)
Net margin			−0.019***	−0.019***	−0.019***	−0.019***
			(0.002)	(0.002)	(0.002)	(0.002)
Liabilities			0.066***	0.066***	0.069***	0.069***
			(0.006)	(0.006)	(0.006)	(0.006)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	No	No	Yes	Yes
Number of observations	2,248,514	2,248,514	2,248,514	2,248,514	2,248,514	2,248,514
R-squared	.71	.71	.71	.71	.71	.71

Table 4, Panel A: eligible supplier dummy

Table 4
Continued

B. Share of eligible suppliers

	(1)	(2)	(3)	(4)	(5)	(6)
Eligible suppliers share × Post	0.069**		0.067**		0.051*	
	(0.030)		(0.029)		(0.029)	
Eligible suppliers share × 2014		0.051		0.043		0.040
		(0.045)		(0.045)		(0.045)
Eligible suppliers share × 2015		0.039		0.035		0.036
		(0.041)		(0.040)		(0.040)
Eligible suppliers share × 2016		0.050**		0.042*		0.027
		(0.025)		(0.025)		(0.025)
Eligible suppliers share × 2017		0.159***		0.155**		0.138**
		(0.062)		(0.060)		(0.060)
log(Assets)			−0.045***	−0.045***	−0.046***	−0.046***
			(0.002)	(0.002)	(0.002)	(0.002)
Cash			0.022***	0.022***	0.026***	0.026***
			(0.005)	(0.005)	(0.005)	(0.005)
PPE			−0.048***	−0.048***	−0.047***	−0.047***
			(0.008)	(0.008)	(0.008)	(0.008)
Net margin			−0.019***	−0.019***	−0.019***	−0.019***
			(0.002)	(0.002)	(0.002)	(0.002)
Liabilities			0.066***	0.066***	0.069***	0.069***
			(0.006)	(0.006)	(0.006)	(0.006)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	No	No	Yes	Yes
Number of observations	2,248,514	2,248,514	2,248,514	2,248,514	2,248,514	2,248,514
R-squared	.71	.71	.71	.71	.71	.71

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts payable to sales (*Accounts payable*). *Has eligible supplier* is a dummy that takes the value of one if a firm had a supplier with CSPP-eligible bonds, and zero otherwise. *Eligible suppliers share* is the number of eligible suppliers divided by the total number of suppliers. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. * $p <$ .1; ** $p <$ .05; *** $p <$ .01.

Table 4, Panel B: share of eligible suppliers

5.6 Table 5 and Table 6: Size controls and matching

Read:

- Table 5 shows that the results survive size decile-year fixed effects.
- The accounts receivable effect remains about 9.5 percentage points, and the accounts payable effect remains about 2.8 percentage points.
- Table 6 uses propensity score matching to improve comparability between treated and control firms.
- The matched-sample estimates still show increases in accounts receivable for eligible firms and accounts payable for customers of eligible firms.

Economic message: The trade-credit channel is not just a size effect. Matching and size-year controls preserve the main results.

Table 5
Size decile-by-year fixed effects

A. Accounts receivable

	(1)	(2)	(3)	(4)
Eligible × Post	0.095***		0.094***	
	(0.032)		(0.032)	
Eligible × 2014		−0.015		−0.013
		(0.022)		(0.022)
Eligible × 2015		0.042		0.045
		(0.043)		(0.043)
Eligible × 2016		0.066*		0.066*
		(0.038)		(0.038)
Eligible × 2017		0.144**		0.144**
		(0.062)		(0.062)
Controls	No	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes
Size decile-by-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	2,248,512	2,248,512	2,248,512	2,248,512
R-squared	.74	.74	.74	.74

B. Accounts payable

	(1)	(2)	(3)	(4)
Has eligible supplier × Post	0.028*		0.028*	
	(0.017)		(0.017)	
Has eligible supplier × 2014		0.019		0.019
		(0.023)		(0.023)
Has eligible supplier × 2015		0.027		0.030
		(0.027)		(0.027)
Has eligible supplier × 2016		0.009		0.011
		(0.016)		(0.017)
Has eligible supplier × 2017		0.082**		0.082**
		(0.034)		(0.034)
Controls	No	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes
Size decile-by-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	2,248,512	2,248,512	2,248,512	2,248,512
R-squared	.71	.71	.71	.71

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts receivable to sales (*Accounts receivable*) and the ratio of accounts payable to sales (*Accounts payable*). *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Has eligible supplier* is a dummy that takes the value of one if a firm had a supplier with CSPP-eligible bonds, and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Firms are sorted into size deciles each year where size is defined as total assets in each year. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 5: Size decile-year fixed effects

Table 6
Propensity score matching

A. Summary statistics (pretreatment)

Eligible firms

	Mean				Median			
	Non treated	Treated	Control	t-test (p-value)	Non treated	Treated	Control	Pearson χ^2 (p-value)
Assets (€million)	24.4	19,890.8	17,066.8	0.808	3.4	9,218.5	9,163.6	0.903
Firm age	21.7	39.2	37.7	0.638	19.0	30.5	33.0	0.807
Net margin	−0.051	0.256	0.186	0.661	0.018	0.181	0.058	0.000
Sales growth	0.177	0.237	0.186	0.676	0.036	0.029	0.027	0.714

Customers of eligible firms

	Mean				Median			
	Non treated	Treated	Control	t-test (p-value)	Non treated	Treated	Control	Pearson χ^2 (p-value)
Assets (€million)	23.8	10,195.2	8,952.2	0.969	3.4	1,920.3	1,906.4	1.000
Firm age	21.7	34.4	29.7	0.018	19.0	26.0	23.0	0.242
Net margin	−0.051	0.062	0.046	0.904	0.018	0.078	0.049	0.242
Sales growth	0.177	0.155	0.231	0.324	0.036	0.047	0.023	0.181

B. Difference-in-differences estimates

	Accounts receivable		*Accounts payable*	
	(1)	(2)	(3)	(4)
Eligible × Post	0.103	0.129*		
	(0.069)	(0.075)		
Has eligible supplier × Post			0.069**	0.059**
			(0.030)	(0.027)
Controls	No	Yes	No	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	1,309	1,272	2,699	2,623
R-squared	.72	.75	.66	.72

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts receivable to sales (*Accounts receivable*) and accounts payable to sales (*Accounts payable*). Panel A shows pretreatment (CSPP announcement) means and medians of nontreated, treated, and control groups and tests of the difference in mean and median between treated and control groups. Treated firms consist of either firms with eligible bonds (eligible firms) or firms with eligible suppliers (customers of eligible firms). The samples includes only treated firms with nonmissing information in Orbis on the year before the treatment. Nontreated firms are all other firms. Control firms are firms that best match treated firms (nearest neighbor) using propensity score matching with replacement on *log*(*Assets*), *Firm age*, *Net margin*, *Sales growth*, and region (exact match on core or periphery countries). Panel B shows estimates of the difference-in-differences regressions. *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Has eligible supplier* is a dummy that takes the value of one if a firm had a supplier with CSPP-eligible bonds, and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of a matched sample based on Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 6: Propensity score matching

5.7 Table 7: Financial constraints of customers

Read:

- The increase in accounts payable is stronger for non-investment-grade and unrated customers than for investment-grade or rated customers.
- It is also stronger for young firms, high-liability firms, and firms with low tangible assets.
- The split-sample results point toward financially constrained firms being more likely to benefit from the trade-credit channel.

Economic message: Trade credit redistributes unconventional monetary stimulus toward customers that face more limited access to external finance.

5.8 Table 8: Core versus periphery countries

Read:

- Eligible suppliers in core countries increase accounts receivable by about 12.6 percentage points.
- Customers in periphery countries with eligible suppliers experience a larger increase in accounts payable, around 7.2 percentage points.
- The effect is not symmetric: liquidity originates mainly in the core but can propagate to the periphery through supplier relationships.

Table 7
Effect of CSPP on accounts payable of eligible firms' customers: The role of financial constraints

Panel A

	Investment-grade (1)	*Non-investment-grade* (2)	*Rated* (3)	*Unrated* (4)	*Mature firms* (5)	*Young firms* (6)	*Low liabilities* (7)	*High liabilities* (8)	*High PPE* (9)	*Low Low PPE* (10)
Has eligible supplier × Post	−0.049	0.041**	−0.055	0.047**	0.018	0.080*	0.023	0.082***	0.022**	0.039*
	(0.048)	(0.021)	(0.038)	(0.022)	(0.015)	(0.042)	(0.020)	(0.031)	(0.010)	(0.023)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	697	2,247,817	1,169	2,247,345	1,153,984	1,086,350	1,124,258	1,124,256	1,124,256	1,124,258
R-squared	.77	.71	.68	.71	.70	.73	.71	.72	.72	.72
Difference (*p*-value)	.07		.02		.17		.11		.51	

Panel B

	High sales (1)	*Low sales* (2)	*Low sales growth* (3)	*High sales growth* (4)	*Low asset growth* (5)	*High asset growth* (6)	*High EBITDA* (7)	*Low EBITDA* (8)	*Public firms* (9)	*Private firms* (10)
Has eligible supplier × Post	0.019	0.191	0.027	0.044**	0.014	0.052**	0.014**	0.066***	0.010	0.073**
	(0.015)	(0.371)	(0.032)	(0.019)	(0.029)	(0.026)	(0.006)	(0.023)	(0.023)	(0.035)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,124,256	1,124,258	1,081,305	1,081,303	1,090,815	1,090,812	1,082,737	1,082,737	10,051	2,238,463
R-squared	.68	.72	.76	.75	.75	.73	.71	.72	.69	.71
Difference (*p*-value)	.75		.63		.47		.06		.13	

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts payable to sales. *Has eligible supplier* is a dummy that takes the value of one if a firm had a supplier with CSPP-eligible bonds, and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Columns 1 and 2, panel A, present the results for the group of firms with investment-grade rating and the group of firms either with a speculative-grade rating or without a credit rating. Columns 3 and 4, panel A, present the results for the group of firms with a credit rating and the group of firms without a credit rating. In columns 5 and 6, panel A, the mature and young firms' groups consist of those firms that are above and below the median of the distribution of firm's age since the year of incorporation. In columns 7 and 8, panel A, the low and high liabilities groups consist of those firms that are below or above the median of the distribution of the ratio of liabilities to assets. In columns 9 and 10, panel A, the low and high PPE groups consist of those firms that are below or above the median of the distribution of the ratio of PPE to assets. In columns 1 and 2, panel B, the low and high sales groups consist of those firms that are below or above the median of the distribution of sales. In columns 3 and 4, panel B, the low and high sales growth groups consist of those firms that are below or above the median of the distribution of sales growth. In columns 5 and 6, panel B, the low and high asset growth groups consist of those firms that are below or above the median of the distribution of asset growth. In columns 7 and 8, panel B, the low and high EBITDA groups consist of those firms that are below or above the median of the distribution of EBITDA-to-assets ratio. Columns 9 and 10, panel B, present the results for the group of publicly listed firms and the group of privately held firms. The difference is the *p*-value of the F-test for the difference on the *Has eligible supplier × Post* coefficient between groups. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

Table 8
Core versus periphery countries

A. Core versus periphery countries

	Core countries		Periphery countries	
	Accounts receivable (1)	*Accounts payable* (2)	*Accounts receivable* (3)	*Accounts payable* (4)
Eligible × Post	0.126***		0.018	
	(0.045)		(0.025)	
Has eligible supplier × Post		0.017		0.072**
		(0.019)		(0.032)
Controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes
Number of observations	650,691	650,691	1,597,823	1,597,823
R-squared	.77	.71	.73	.71

B. Core versus periphery countries eligible suppliers

	Euro area countries	Core countries	Periphery countries
	Accounts payable		
	(1)	(2)	(3)
Has core eligible supplier × Post	0.044**	0.018	0.077
	(0.020)	(0.020)	(0.051)
Has periphery eligible supplier × Post	0.024*	0.003	0.034*
	(0.013)	(0.023)	(0.018)
Controls	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes
Number of observations	2,248,514	650,691	1,597,823
R-squared	.71	.71	.71

This table presents difference-in-differences estimates of firm-level panel regressions of the ratio of accounts receivable to sales and the ratio of accounts payable to sales. *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Has eligible supplier* is a dummy that takes the value of one if a firm had a supplier with CSPP-eligible bonds, and zero otherwise. *Has core eligible supplier* is a dummy variable that takes the value of one if a firm is a customer of eligible firms with headquarters in core countries, and zero otherwise. *Has periphery eligible supplier* is a dummy variable that takes the value of one if a firm is a customer of eligible firms with headquarters in periphery countries, and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

Economic message: Production networks allow the benefits of a core-centered corporate bond purchase program to reach periphery firms.

5.9 Tables 9 and 10: Real effects for eligible firms and customers

Read:

- Table 9 shows that eligible firms increase asset growth and accounts receivable after the CSPP. Their accounts payable also increases, but the main supplier-side trade-credit effect is through receivables.
- Table 10 shows that customers of eligible firms increase asset growth, capital expenditures, inventories, accounts receivable, and labor growth.
- Customer employment growth rises by about 3.4 percentage points in the baseline customer real-effects table.
- Customers also increase accounts payable, consistent with receiving more trade credit financing.

Economic message: Trade credit affects real activity. Customers use additional financing to expand investment, working capital, and employment.

Table 9
Real effects of CSPP: Investment and financing of eligible firms

A. Investment in fixed assets, working capital, and human capital

	Asset growth (1)	*CAPEX* (2)	*ΔInventories* (3)	*ΔAccounts receivable* (4)	*Labor growth* (5)
Eligible × Post	0.023*	0.002	0.005	0.009**	0.018
	(0.014)	(0.005)	(0.004)	(0.004)	(0.024)
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	2,248,512	2,116,731	2,246,443	2,247,599	1,779,908
R-squared	.55	.48	.32	.29	.28

B. External and internal financing

	ΔTotal debt (1)	*ΔLong-term debt* (2)	*ΔShort-term debt* (3)	*ΔAccounts payable* (4)	*ΔCash* (5)
Eligible × Post	0.016	0.009	0.005	0.008**	0.003
	(0.011)	(0.011)	(0.006)	(0.003)	(0.005)
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	2,081,233	2,090,098	2,236,327	2,238,452	2,231,250
R-squared	.26	.22	.18	.27	.43

This table presents difference-in-differences estimates of firm-level panel regressions of eligible firm outcomes. Panel A shows regressions for asset growth, CAPEX (scaled by lagged assets), change in inventories (scaled by lagged assets), change in receivable (scaled by lagged assets), and labor growth. Panel B shows regressions for change in total debt, change in long-term debt, change in short-term debt, change in accounts payable, and change in cash (all variables scaled by lagged assets). *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 9: Real effects for eligible firms

Table 10
Real effects of CSPP: Investment and financing of eligible firms' customers

A. Investment in fixed assets, working capital, and human capital

	Asset growth (1)	*CAPEX* (2)	*ΔInventories* (3)	*ΔAccounts receivable* (4)	*Labor growth* (5)
Has eligible supplier × Post	0.025*	0.005*	0.004**	0.011**	0.034*
	(0.015)	(0.003)	(0.002)	(0.005)	(0.020)
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	2,248,512	2,116,731	2,246,443	2,247,599	1,779,908
R-squared	.55	.48	.32	.29	.28

B. External and internal financing

	ΔTotal debt (1)	*ΔLong-term debt* (2)	*ΔShort-term debt* (3)	*ΔAccounts payable* (4)	*ΔCash* (5)
Has eligible supplier × Post	0.008	0.005	0.004	0.009***	−0.007
	(0.007)	(0.007)	(0.004)	(0.003)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	2,081,233	2,090,098	2,236,327	2,238,452	2,231,250
R-squared	.26	.22	.18	.27	.43

This table presents difference-in-differences estimates of firm-level panel regressions of eligible firms' customers outcomes. Panel A shows regressions for asset growth, CAPEX (scaled by lagged assets), change in inventories (scaled by lagged assets), change in receivable (scaled by lagged assets), and labor growth. Panel B shows regressions for change in total debt, change in long-term debt, change in short-term debt, change in accounts payable, and change in cash (all variables scaled by lagged assets). *Has eligible supplier* is a dummy that takes the value of one if a firm had a supplier with CSPP-eligible bonds, and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 10: Real effects for customers

5.10 Tables 11 and 12: Customer relationships and market share

Read:

- Table 11 shows that eligible firms keep more existing customers and initiate more new customer relationships after the CSPP.
- The increase is concentrated in core countries: core eligible firms keep about 11.7 more customers and add about 4.25 new customers in the simplest specification.
- Periphery eligible firms do not show the same significant increase.
- Table 12 shows that eligible firms increase sales market share at the four-digit SIC level in the euro area and especially in core countries.

Economic message: The CSPP may have competitive consequences. Firms directly benefiting from bond purchases can strengthen their upstream position by using trade credit to retain and attract customers.

Table 11
Effect of CSPP on customer relationships maintained and new relationships initiated of eligible firms

	Number of customers kept			Number of new customers		
	(1)	(2)	(3)	(4)	(5)	(6)
A. Euro area countries						
Eligible × *Post*	9.514***	8.223***	7.624***	3.281***	2.767**	2.715**
	(1.935)	(2.073)	(2.043)	(1.234)	(1.231)	(1.219)
Controls	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Number of observations	9,434	6,045	6,037	9,434	6,045	6,037
R-squared	.84	.84	.84	.58	.57	.57
B. Core countries						
Eligible × *Post*	11.700***	9.968***	9.729***	4.247***	3.429**	3.612**
	(2.334)	(2.510)	(2.481)	(1.482)	(1.501)	(1.471)
Controls	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Number of observations	6,558	3,883	3,883	6,558	3,883	3,883
R-squared	.84	.84	.84	.59	.58	.58
C. Periphery countries						
Eligible × *Post*	0.624	−0.359	−0.580	−0.419	0.275	−0.648
	(1.318)	(1.445)	(1.548)	(1.152)	(1.058)	(1.248)
Controls	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	No	Yes	No	No	Yes
Number of observations	2,876	2,162	2,154	2,876	2,162	2,154
R-squared	.85	.85	.85	.51	.50	.52

This table presents difference-in-differences estimates of firm-level panel regressions of the number of customer relationships maintained and number of new customer relationships initiated. The dependent variable in columns 1–3 is the number of customer relationships maintained by a supplier relative to the existing relationships in the previous year. The dependent variable in columns 4–6 is the number of new customer relationships initiated by a supplier relative to the existing relationships in the previous year. *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of FactSet Revere Supply Chain Relationship nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 11: Customer relationships

Table 12
Effect of CSPP on sales market share of eligible firms

	Euro area countries		Core countries		Periphery countries	
	(1)	(2)	(3)	(4)	(5)	(6)
Eligible × *Post*	0.373*	0.374*	0.499*	0.499*	0.108	0.109
	(0.226)	(0.226)	(0.302)	(0.302)	(0.297)	(0.297)
Controls	No	Yes	No	Yes	No	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	2,248,514	2,248,514	650,691	650,691	1,597,823	1,597,823
R-squared	.98	.98	.98	.98	.97	.97

This table presents difference-in-differences estimates of firm-level panel regressions of sales market share at the four-digit SIC level. *Eligible* is a dummy variable that takes the value of one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement date (March 2016), and zero otherwise. *Post* is a dummy variable that takes the value of one in the years 2016 and 2017, and zero otherwise. Regressions include the same control variables as those in Table 2 (coefficients not shown). The sample consists of Bureau Van Dijk's Orbis nonfinancial firms based in the euro area in the 2013–2017 period. All explanatory variables are lagged by 1 year. Table A.1 in the appendix defines the variables. Robust standard errors adjusted for firm-level clustering are reported in parentheses. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 12: Sales market share

5.11 Appendix Table A.1: Variable definitions

Read:

- Table A.1 defines the main treatment variables, network variables, trade-credit outcomes, controls, and real outcomes.
- It is useful for interpreting the difference between supplier-side variables such as Eligible and customer-side exposure variables such as Has eligible supplier.

Economic message: The variable definitions clarify that the paper measures both sides of the same financing relationship: suppliers' receivables and customers' payables.

Table A.1
Variable definitions

Variable	Definition
Eligible	Dummy variable that equals one if a firm had bonds eligible for purchase under the CSPP before the CSPP announcement, and zero otherwise
Eligible issuer	Dummy variable that takes the value of one if an eligible firm has issued bonds after the CSPP announcement date, and zero otherwise
Eligible nonissuer	Dummy variable that takes the value of one if an eligible firm has not issued bonds after the CSPP announcement date, and zero otherwise
Has eligible supplier	Dummy variable that equals one if a firm is reported as a customer of eligible firms in FactSet Revere Supply Chain Relationships database, and zero otherwise
Eligible suppliers share	Number of eligible suppliers divided by total number of suppliers
Has core eligible supplier	Dummy variable that equals one if a firm is a customer of an eligible firm with headquarters in core countries, and zero otherwise
Has periphery eligible supplier	Dummy variable that equals one if a firm is a customer of an eligible firm with headquarters in periphery countries, and zero otherwise
Accounts receivable	Accounts receivable (Orbis item *DEBTORS*) divided by operating revenue (Orbis item *OPRE*)
Accounts payable	Accounts payable (Orbis item *CREDITORS*) divided by operating revenue (Orbis item *OPRE*)
Assets	Total assets (Orbis item *TOAS*)
Sales	Operating revenue (Orbis item *OPRE*)
Cash	Cash and cash equivalent (Orbis item *CASH*) divided by total assets (Orbis item *TOAS*)
PPE	Tangible fixed assets (Orbis item *TFAS*) divided by total assets (Orbis item *TOAS*)
Net margin	Net income (Orbis item *PL*) divided by operating revenue (Orbis item *OPRE*)
Liabilities	Current liabilities (Orbis item *CULI*) plus noncurrent liabilities (Orbis item *NCLI*) divided by total assets (Orbis item *TOAS*)
Asset growth	Change in total assets (Orbis item *TOAS*) divided by the previous year total assets
CAPEX	Change in tangible fixed assets (Orbis item *TFAS*) plus depreciation and amortization (Orbis item *DEPR*) divided by the previous year total assets (Orbis item *TOAS*)
Labor growth	Change in number of employees (Orbis item *EMPL*) divided by the previous year number of employees
Δ*Inventories*	Change in inventories (Orbis item *STOK*) divided by the previous year total assets (Orbis item *TOAS*)
Δ*Accounts receivable*	Change in accounts receivable (Orbis item *DEBTORS*) divided by the previous year total assets (Orbis item *TOAS*)
Sales growth	Change in operating revenue (Orbis item *OPRE*) divided by the previous year operating revenue
EBITDA	EBITDA (Orbis item *EBTA*) divided by the previous year total assets (Orbis item *TOAS*)
Firm age	Number of years since the year of incorporation (*DATEINC_YEAR*)
Δ*Accounts payable*	Change in accounts payable (Orbis item *CREDITORS*) divided by the previous year total assets (Orbis item *TOAS*)
Δ*Total debt*	Change in financial debt (Orbis item *LTDB* plus Orbis item *LOAN*) divided by the previous year total assets (Orbis item *TOAS*)
Δ*Long-term debt*	Change in long-term debt (Orbis item *LTDB*) divided by the previous year total assets (Orbis item *TOAS*)
Δ*Short-term debt*	Change in current loans (Orbis item *LOAN*) divided by the previous year total assets (Orbis item *TOAS*)
Δ*Cash*	Change in cash and cash equivalents (Orbis item *CASH*) divided by the previous year total assets (Orbis item *TOAS*)

6 Short summary

- The paper studies how the ECB's CSPP transmits through trade credit in production networks.
- Eligible nonfinancial bond issuers increase accounts receivable after the CSPP.
- Customers of eligible suppliers increase accounts payable after the CSPP.
- The results show no strong pretreatment differences and remain robust to controls, fixed effects, matching, and size-year fixed effects.
- Near the investment-grade cutoff, the supplier-side effect is especially large, supporting the eligibility-based interpretation.
- Eligible firms that issue bonds after the CSPP expand trade credit more than eligible nonissuers.
- Financially constrained customers benefit more from the trade-credit channel.
- Core-country eligible suppliers transmit liquidity to periphery-country customers.
- Customers increase investment, inventories, accounts receivable, and employment.
- Eligible firms expand their customer base and increase market share, implying possible concentration effects.
- The core message is that unconventional monetary policy can be transmitted through firm-to-firm credit, not only through banks or public capital markets.

7 Conclusion

7.1 Core conclusion

Adelino, Ferreira, Giannetti, and Pires show that the ECB's corporate bond purchase program affected more than the firms whose bonds were eligible for purchase. Eligible firms expanded trade credit to their customers, and customers then expanded real activity. Production networks therefore transmitted unconventional monetary policy beyond public bond markets.

7.2 Economic intuition

Large eligible firms receive cheaper bond-market financing. Because they sell to other firms and can finance customers through delayed payment terms, they can pass liquidity downstream by extending more trade credit. Customers, especially financially constrained ones, treat this trade credit as a source of financing and respond by increasing investment and employment.

7.3 Takeaway

The paper adds a production-network channel to monetary transmission. It shows that unconventional monetary policy can reach small and constrained firms indirectly through supplier credit. At the same time, the policy can strengthen eligible suppliers' product-market position by allowing them to attract and retain customers.